

Design and Implementation of Student Management System of Educational Management System

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Abstract—The universities educational management System is a typical information management system,including the establishment and maintenance of backstage database and the development of frontstage application program.For the former,it requires a database of data consistency and integrity to build strong,full and good,and for the latter procedure,it requires full-functioned and easy to use and so on. This paper mainly introduces the design and implementation of students educational management system.After research and analysis,the system uses the C # and SQL Server 2000 to develop,with a three-tier,including the system development environment and technical support,the design of system,the implementation of system and concluded.The main function is for all students, providing them with a teaching management platform to facilitate the management of personal information and cooperate a variety of the teaching mission of schools.

Keywords—educational management, school register management, curriculum management

I. INTRODUCTION

A. Significance of topic selection and development background

University educational administration management system is the teaching, educational administration application software. Educational administration is one of the main daily management work of university, in school, department, division, many aspects, with the continuous reform of teaching system, especially, the spread of elective system and credit system, educational administration management work Increasingly heavy and complicated. Thus promoting the modernization of teaching management process. This school is the educational administration system to summarize the data, one of the more important link in management, through the upgrading of educational administration system, can be very handy for the school within the existing each kind of information do a more intuitive understanding, so as to improve the management efficiency, saving a large amount of manpower and material resources [1]. Through the computer network to realize the information sharing, integration of data summary and analysis of the query. As a part of computer application, the use of educational administration information management, has the crafts incomparable advantages. For example:

retrieves, the search rapidly convenient, the reliability high, thereserves big, the secrecy good, the long life and low cost. These advantages can greatly improve the efficiency of school management, school management scientific and standardization management, and the important condition of the world.

B. Research content and the structure of the thesis

1) The paper research content

According to the software life cycle theory [2] to the university educational administration management system is studied. Wide-ranging university educational administration management system, available to system administrators, teachers and students, the university educational administration management system can be divided into administrator subsystem, management subsystem of the teachers and students management subsystem. This paper mainly studies the college student management subsystem design and implementation of educational administration system.

2) Structure of the thesis

Chapter1: introduction.This chapter summarizes the system selected topic significance and the development background, this paper studies the content and structure arrangement.

Chapter2: the system development environment and technical support.This chapter first describes the development of the system language is c # and Visual Studio development environment are introduced, then the system used by the database SQL Server 2000 introduction, finally to use ADO.NET to access the database technology.

Chapter3: the university educational administration management system student management subsystem design.This chapter first on the system function module division, draw the system hierarchy structure, and then designs a database model of system.

Chapter4: This chapter mainly introduced the system each module programming implementation, focus on the system management module, student status management module programming implementation has carried on the detailed introduction.

Chapter5:This chapter of the whole system design and implementation process are summarized.

II. SYSTEM DEVELOPMENT ENVIRONMENT AND TECHNICAL SUPPORT

A. Summary of c # and visual studio

C # is Microsoft launched a new, object oriented programming language. It allows the programmer to write quickly various applications based on Microsoft.NET platform, Microsoft.NET provides a series of tools and services to maximize the development and utilization of the computing and communications.

C # main characteristics: simple, modern, object-oriented, type safety [3]. Visual Studio is a complete set of development tools, used to generate the ASP.NET Web applications, XML Web services and mobile applications, desktop applications. Visual Basic, Visual c # and Visual c + +, all use the same integrated development environment (IDE), such tools Shared could be carried out, and be able to easily create mixed language solutions. In addition, the language use. The function of the .net Framework, it provides can simplify the ASP Web applications and the key technology of XML Web services development [4].

It's important to note that do not always need to Visual Studio to the preparation of the program. In fact can be downloaded by the .net SDK for c # application development, the .net SDK contains the c # compiler, but because the compiler is based on the order form, it more difficult for the large application development, so developers generally choose Microsoft Visual Studio development tools for the development of large application. Visual Studio development is a powerful tool that can be used to develop a variety of applications, the support of the language is also very rich, is by far the best tool used to develop c # applications. This paper version is used in the Visual Studio 2005 [5].

B. SQL Server 2000

SQL Server 2000 is Microsoft's version of SQL Server database management system. This version inherits the advantage of the SQL Server version 7.0, at the same time than it adds many more advanced features, easy to use, good scalability, and related software higher integration degree can span from running Microsoft Windows 98 laptop to run Microsoft Windows 2000 large multiprocessor servers using a variety of platforms.

Common database Access, Microsoft SQL Server, Oracle, IBM DB2, Sybase, etc., including SQL Server is the applicability and efficiency combined with one of the best database. On the network at present Web applications use more database is SQL Server. This article USES the SQL Server 2000 personal edition.

C. Use ADO.NET to access the database

Active Data Objects (ADO) was developed by Microsoft object-oriented Data access library [7], has been widely used. ADO.NET is the follow-up technology of ADO, but is not simple to upgrade, but has a very big improvement.

Using ADO.NET, the programmer can be very simple and fast access to the database.

No matter from the point of grammar, or from the point of style and design target, ADO.NET and ADO are significantly different. To access to database through ADO in ASP, typically through the following four steps:

- to create a link to the database, namely ADO. Connection.
- query a data set that execute SQL, create a you.
- to the operation of the data collection needs.
- close the data link.

In ADO.NET, has changed a lot these steps. ADO.NET is one of the most important concept of the DataSet. The DataSet is not dependent on the database of data sets independently. So-called independence, is: even if disconnect data link, or close the database and the DataSet is still available. If you used the connection records in ASP collection (Connectionless you), then the DataSet is the technology of the most radical alternatives. With the DataSet, so, ADO.NET to access the database correspondingly changed the step:

- create a database link.
- to request a record collection.
- to record collection temporary to the DataSet.
- if need be, return to step 2.
- close the database link.
- needed to do operation on the DataSet.

The DataSet is using XML to describe data internally. Because XML is a platform independent, unrelated language data description language, and can describe complex data relationships of data, such as parent-child relationships of data, so the DataSet can actually hold with complex relationships of data, and no longer dependent on the database link.

III. THE DESIGN OF THE UNIVERSITY STUDENT EDUCATIONAL ADMINISTRATION SYSTEM MANAGEMENT SUBSYSTEM

A. Description of system function module

University educational administration management system student management subsystem design goal is convenient student users. Through the application, the student can log on to enter the system and a series of related operations. Students end main modules as follows [6].

a) the system management

Landing: college students through the window of landing to land.

Cancellation: students complete landing after operation can cancel their own behavior. After the completion of the exit system: all operating directly from the system.

b) the registration

New registration: the first students should submit their personal information to the system, namely the

registration. Modify information: after landing students can modify their personal information.

c) the course management

All course list: display all courses in all of the teachers.

My course list: display all courses currently taking.

Add classes: taking courses in the teacher.

Delete the course selection: delete have elective course.

d) management band 4 and band 6

Enter oneself for an examination to enter oneself for an examination the cet: add information. Enter oneself for an examination history: list all previous cet to enter oneself for an examination records.

e) computer grade examination management

Enter oneself for an examination to enter oneself for an examination grade exam: add information. Enter oneself for an examination history: list all previous computer grade examination records.

B. The system level diagram

Hierarchical graph to describe the software hierarchy, hierarchy figure is very suitable for used in the process of top-down design software.

C. Database design—the concept of database structure design

In the educational administration system student management subsystem, using SQL Server 2000 database, used to store the student information, course information, student course selection information, cet management information and management information computer grade examination, etc.

Through the system requirements analysis and the determination of system function structure, planning the system used in the database entity objects and entities e-r diagram.

Student management system is the main user of the students, thus requiring students entities, including student student id, password, name, admission time, etc.

Students taking object is course, so you need course entities, including course number, course name, class time attribute, etc.

Manage the cet, need cet management information entities, including the number, enter oneself for an examination student student id, exam grades and other properties.

Student course selection and management of computer grade examination information entity and e-r diagram design similar to the above three entities, not repeat here.

IV. THE REALIZATION OF THE UNIVERSITY STUDENT EDUCATIONAL ADMINISTRATION SYSTEM MANAGEMENT SUBSYSTEM

A. System management module programming

Student users want to want to achieve the function of student management system module, must have a system to run the main interface, this is the student in the presentation layer framework of the project.

1) the students landing

Students can open the system main interface first, and then select the main interface of "system management" | "student login" command, the pop-up dialog box shown. Enter student id and password after you click "ok" button to enter the system, if the input information is correct, then log in successfully, the main interface of all the menu will be enabled, if the land is not successful, may pop up error message dialog.

2) the student logout

To realize "student logout" function, simply click students "logout" under "system management" in the options

3) log out

Implementation of "exit system" function, click on the "system management" under the same "log out" option.

B. Student status management module programming

1) the new registration

New registration system to provide a form, namely the students can through the form input your personal information and make complete registration database. The realization of the "new registration" function, is to choose the main interface menu bar on the "registration" | "admissions," popup indicating the form system. If the input information is legitimate, click the "submit registration" button to complete registration operation; If the input information is not correct, error message will appear.

2) modify the information

To implement this feature, first you need to specify the student's information from the database records, and then through the form, the form can be used to input the new student information at the same time. When information modification can be finished through the form to update the revised information to the database. The function of the interface and implementation similar to register, not repeat here. After landing, the choice under the "management" of the main interface of the modified information.

C. Course management module programming

1) all the course list and add classes

Implement this function, simply select the interface on the menu bar "curriculum management" | "all course list,"

2) the course list and delete my course

"My course list" and "whole course list" basic similar, the difference is that the former will of course information while reading the database matching courses for the current user records to be read. The difference is different in the

picture above "choose the course" button to "give up the course", the button can delete the course selection function.

D. The realization of the cet management module

1) the graded English test to enter oneself for an examination history

After log in successfully, choose the main interface menu "cet management" | "to enter oneself for an examination history,"

2) enter oneself for an examination the cet

After log in successfully, choose the main interface menu bar on the "management of band 4 and band 6" | "to enter oneself for an examination the cet," pop-up interface, as shown in figure 4.7 the user input information after click the "submit" button, the application will check the validity of legal information will be submitted to the database. The results will be returned in the form of dialog,

E. The realization of computer grade examination management module

After finish level test band 4 and band 6 management module, due to the development of the computer grade examination way similar to the cet management, there is no longer here.

REFERENCES

- [1] L. Xu, S.Hui, Information systems planning method [J]. Journal of shanxi university, 2002, 32 (4) : 42 -45.
- [2] Hans Van Vliet. Software Engineering - all of the and Practice [M]. Second Edition. New York: John Wiley & Sons, 2000:120-178.
- [3] X.Song, L.Zhao, C # easy introduction[M]. Beijing: people's posts and telecommunications press, 2009:1-131.
- [4] Y.Huang, Research and development enterprise Web information system [D]. xi 'an university of science and technology, 2003.
- [5] Bauer, snell. Visual Studio 2005 techniques [M]. people's posts and telecommunications press, 2008:441-527.
- [6] Y. Yu, B/S architecture in the design and implementation of educational administration management system [D]. dalian maritime university, 2000.